EMNLP 2026

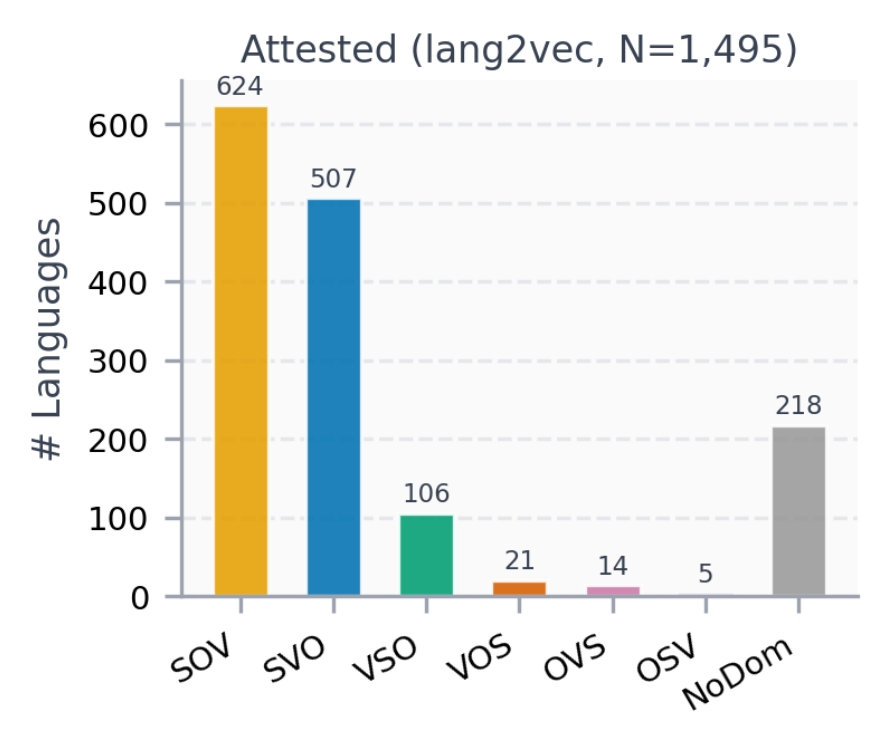
Left-Branching Transformers Excel at Right-Branching Languages

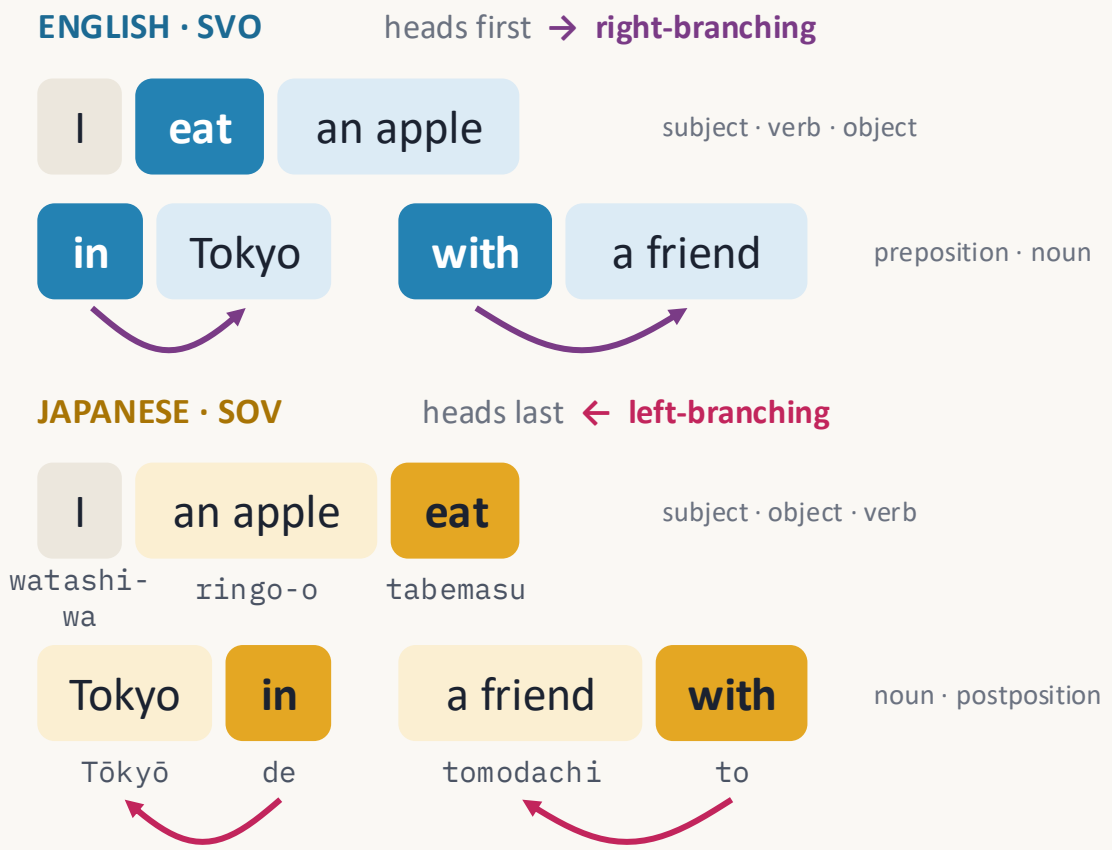
Data Shapes Word Order Preferences in Language Models

Varvara Arzt · Allan Hanbury · Terra Blevins

TU Wien · Northeastern University

Natural Languages Vary a Lot in their Word Order





Do LMs Handle All Word Orders Equally Well?

RESEARCH QUESTIONS

RQ1 Do LMs have systematic word order preferences?

RQ2 If there are systematic preferences, are these architectural or data-driven?

RQ3 Do LM preferences mirror attested patterns?

RQ4 Do LM preferences on artificial languages match those on natural languages?

Prior work: mixed results, partly because each study looks at only **one setting**

White & Cotterell 2021 · Kallini et al. 2024 · Kuribayashi et al. 2024 · El-Naggar et al. 2025 · Ziv et al. 2026

Artificial languages only

Controlled, but no semantics, morphology or discourse: findings may not transfer

Natural languages only

Realistic, but word order is confounded with everything else

→ We do both: same architecture, matched training data scales

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Experimental Setup

Natural languages

Languages

>70 typologically diverse languages

Same language set at every data size

Models

Monolingual **Goldfish** (GPT-2 architecture)

+ multilingual mGPT, XGLM, BLOOM

Training data

Web text: word order + everything else

5 MB → 1 GB per language, byte-premium scaled

Evaluation

FLORES-200 & PUD

BPEC (bits per English character)

Shared across b

$$\mathrm{BPEC}_\ell = \frac{-\sum_{s \in S_\ell} \sum_{t=1}^{n_s} \log_2 p_\theta\left(x_t^{(s)} \mid x_{<t}^{(s)}\right)}{C_{\mathrm{eng}}}$$

BPEC: bits per English character

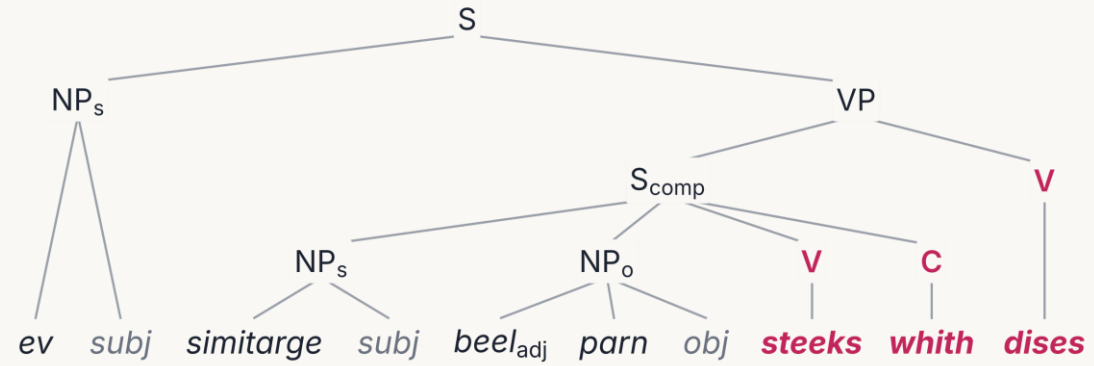
NLL ÷ English character count: comparable across tokenisers

Building Artificial languages

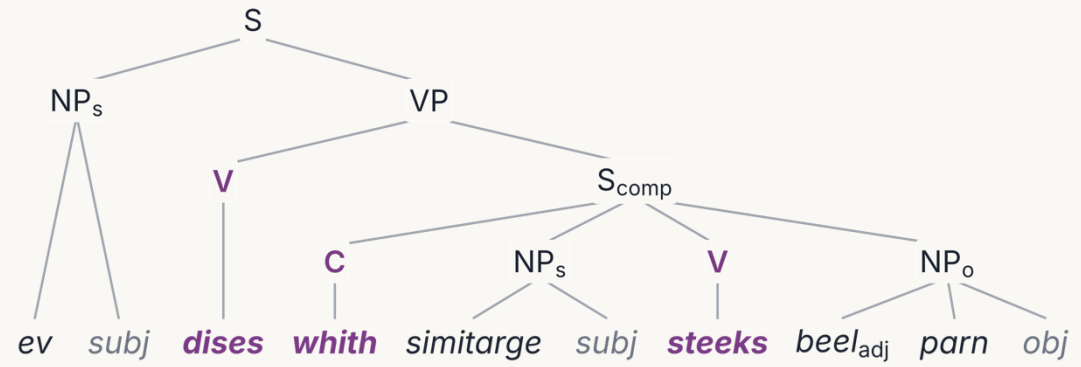
Switch	L	R
1 **Complement clause**	S-comp V	V S-comp
2 **Complementiser**	S Comp	Comp S
3 **Adposition**	NP Postp	Prep NP
4 **Adjective**	Adj N	N Adj
5 **Relative clause**	VP Rel N	N Rel VP

6 base orders × 2^5 = 192 languages

SOV-LLLLL ~ Japanese



SVO-RRRLR ~ English



Labeling Natural Languages the Same Way

ONE LABEL PER LANGUAGE

SVO- **R** **R** **R** **L** **R**

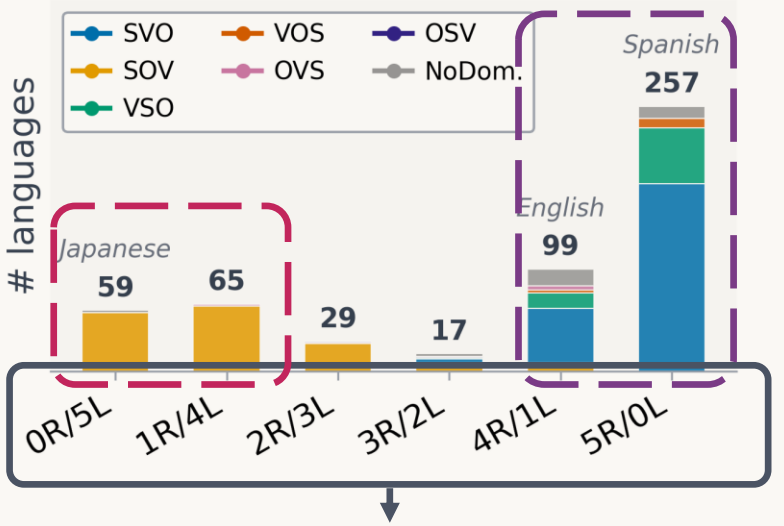
1 complement clause · **2** comple-mentiser · **3** adposition · **4** adjective · **5** relative clause

base word order + 5 binary switches: **L** left- / **R** right-branching

Language	Label	Scale
English	SVO-**RRRLR**	4R/1L
French	SVO-**RRRRR**	5R/0L
Hindi	SOV-**LRLLL**	1R/4L
Japanese	SOV-**LLLLL**	0R/5L

Labels from WALS, Grambank and APiCS; only languages with all five switches annotated

Documented natural language distribution (N=526)



Scale = number of right-branching switches

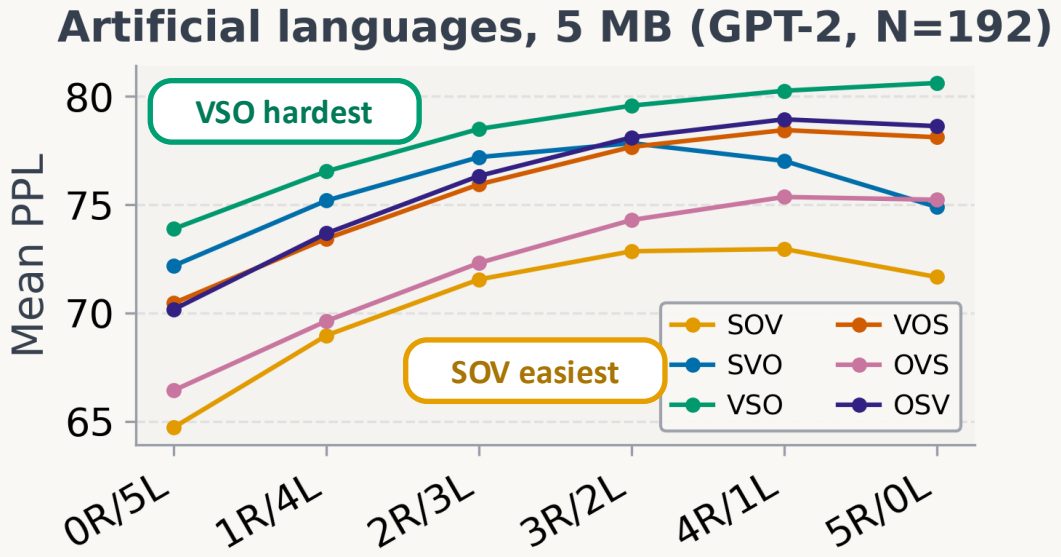
0R/5L fully left-branching · **5R/0L** fully right-branching

Natural languages cluster at both extremes: **mostly left-** and **mostly right-branching**

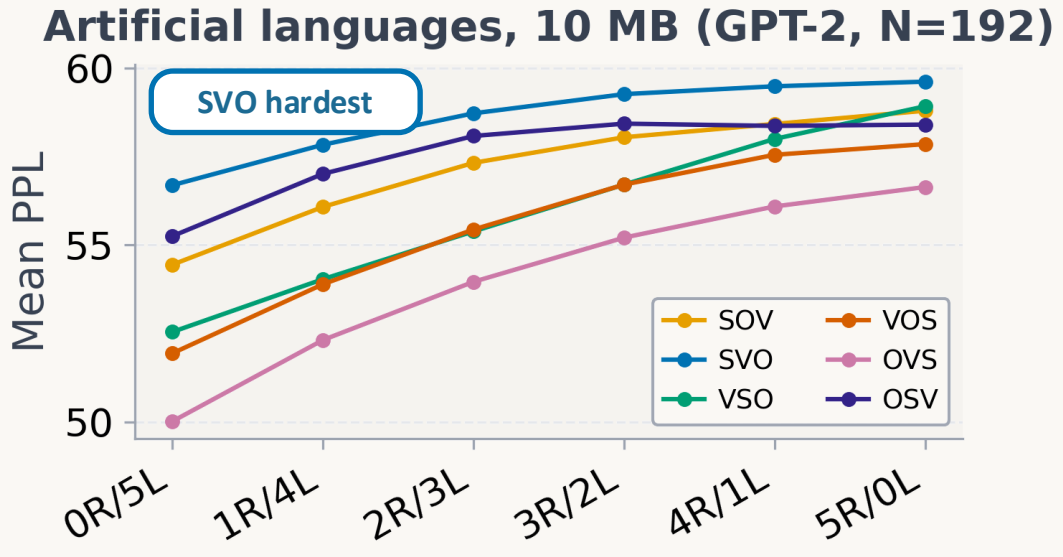
LM Preferences in Artificial Languages

Base word order rankings flip with more training data

GPT-2 · 5 MB



GPT-2 · 10 MB



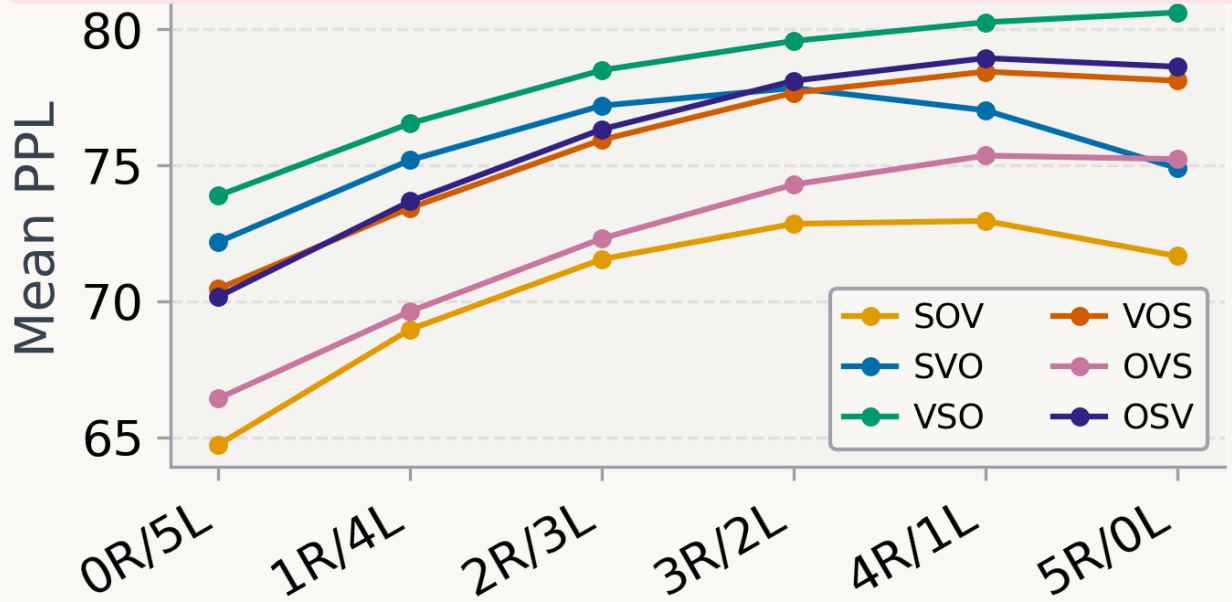
5 MB: **SOV** easiest, **VSO** hardest → **10 MB:** VS orders easiest, **SVO** **hardest**

No stable base word order preference; at scale on natural languages, **SVO** will be the **easiest**

LM Preferences in Artificial Languages

Left-branching is easiest for all six base word orders, consistently at 5 and 10 MB

Left-branching preference also reported in prior work
White & Cotterell, 2021; Kuribayashi et al., 2024; El-Naggar et al., 2025

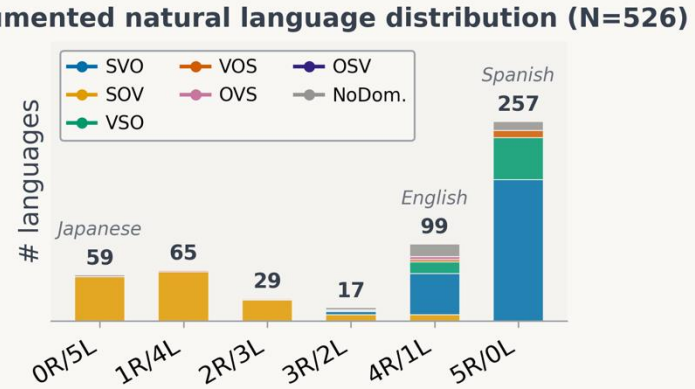


← fully left-branching
e.g. SOV-LLLLL ≈ Japanese

fully right-branching →
e.g. SVO-RRRRR ≈ French

Mean PPL over 10 runs · lower = easier

Natural languages: SOV mostly left-, SVO mostly right-branching

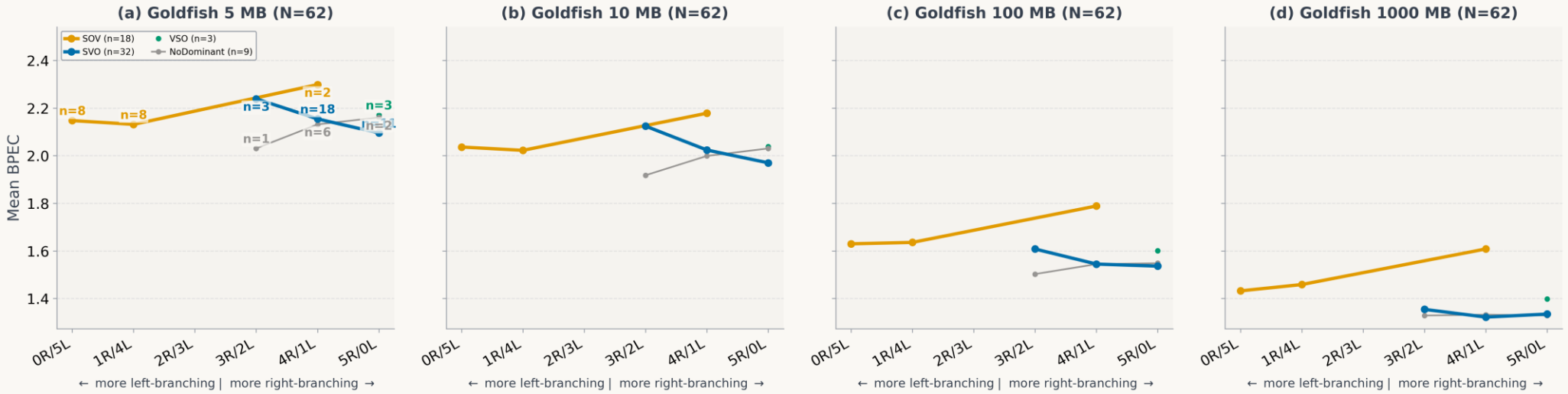


LMs show no such pattern on artificial languages: left-branching is easiest for every base word order

The easiest configuration for each base word order has **no natural counterpart**

LM Preferences in Natural Languages

No preference at 5 MB. With more data, right-branching SVO becomes **easiest**.



	(a) 5 MB	(b) 10 MB	(c) 100 MB	(d) 1000 MB
SVO–SOV δ	0.00	−0.10	**−0.60*** **	**−0.76*** **

Opposite of artificial languages: there, SVO is the **hardest** base word order at 10 MB; here, the **easiest** at scale

Same languages at every size: by 1 GB, SOV is ≈10% harder to model, though it is the most frequent order worldwide

Also holds for multilingual models (XGLM, mGPT) · details in the paper

Mean BPEC on FLORES-200, lower = easier · δ = Cliff's δ, < 0: SVO easier · *** p < .001

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What Drives the SVO Advantage?

Linear mixed-effects model: the growing SVO–SOV gap survives every control

Not script. 71% of SOV vs 19% of SVO FLORES-200 languages are non-alphabetic, but on alphabetic scripts only the gap still grows with data

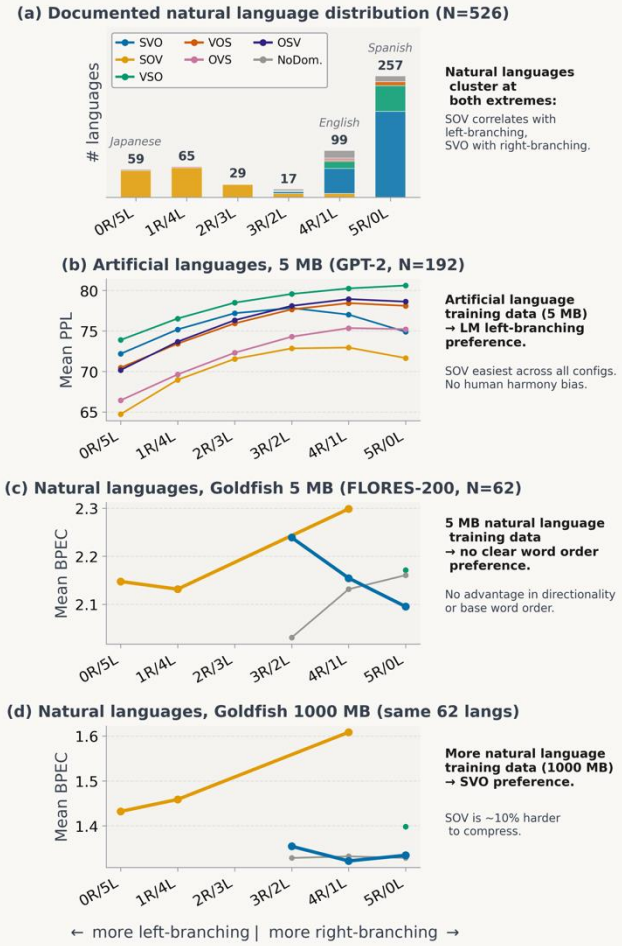
Not morphological complexity. Its effect shrinks with data, so it cannot explain a gap that grows

Not training data composition. Web-crawl (OSCAR) vs. curated share of each language's data: no effect on the growth

Partly language resourcedness (≈20% of the effect): a proxy for data quality

Linear mixed-effects model, 50 SVO/SOV languages × 4 sizes: BPEC ~ word order × size + covariates + (1 | language)

Takeaways



1 **Systematic, but not human-like: left-branching** on **artificial languages** regardless of base word order and against word order correlations in **natural languages**

2 **Data overrides the architectural bias:** on **natural languages** at scale, better-resourced, **right-branching SVO** languages become easiest, in mono- and multilingual models

3 **LM preferences on artificial and natural languages go in opposite directions at scale: SVO** is hardest on **artificial languages**, easiest on **natural ones**

Even monolingual models trained on just 1 GB develop the **SVO** preference; frontier models, with far larger resource gaps between languages, likely amplify it.

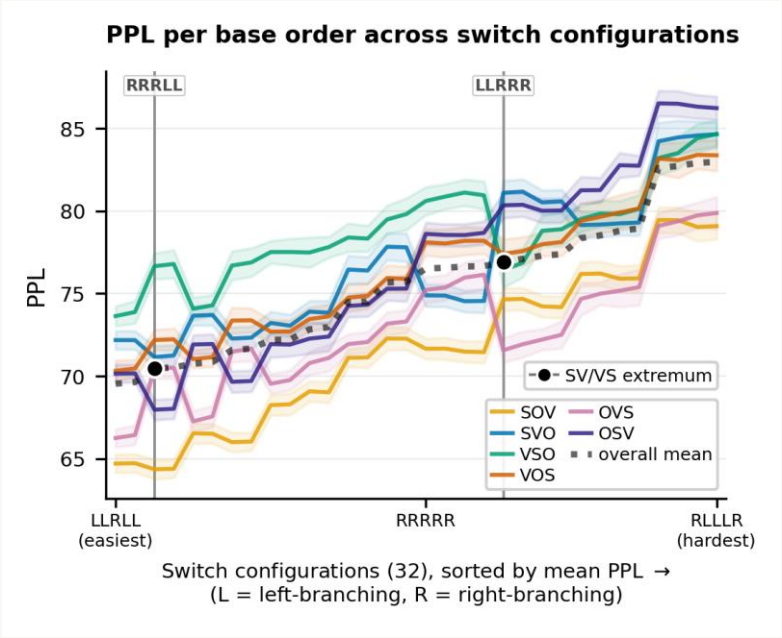
A precondition for ***silent typological homogenisation***: flexible word order languages drifting towards the dominant ones.

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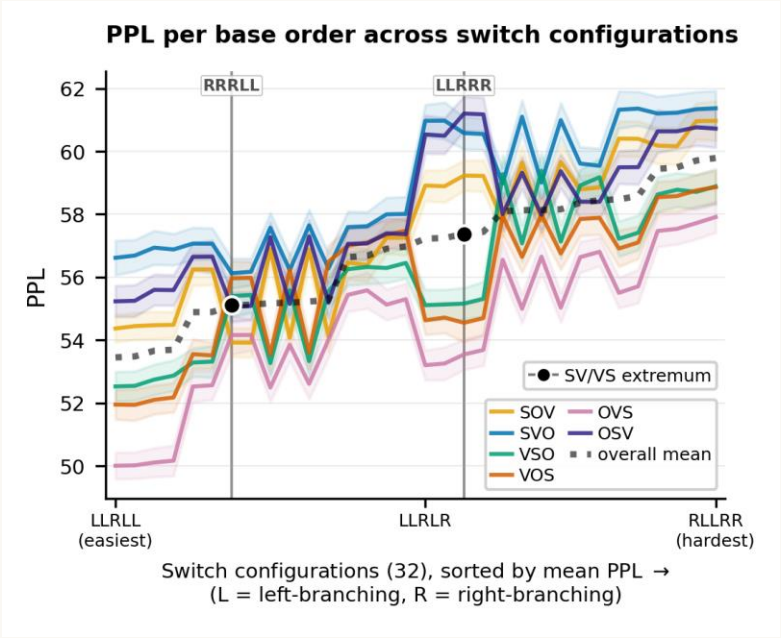
LM Preferences in Artificial Languages

Relative clauses drive left-branching; base word order rankings flip with data

GPT-2 · 5 MB



GPT-2 · 10 MB



Internal constituent order, not base word order, drives the spread: all six lines rise together

Relative clauses matter most: prenominal ones are 4–7 PPL easier, yet ~4× rarer in natural languages

5 MB: SOV easiest, VSO hardest
10 MB: VS orders easiest, **SVO hardest**

All 32 switch configurations, sorted by mean PPL (left = easiest, L = left-, R = right-branching); one line per base word order. GPT-2, mean over 10 runs.

Multilingual Models: the Same SVO Advantage

Model	SVO	SOV	NoDom	δ
BLOOM-7b1	2.81 (2.06)	**1.82** (.29)	—	+.25
XGLM-1.7B	**1.41** (.88)	1.78 (5.12)	1.42 (.54)	**−.53***
mGPT-13B	**1.44** (.25)	1.68 (.18)	1.54 (.24)	**−.44***

Median FLORES-200 BPEC (IQR); ↓ better, bold = best per model. δ = SVO–SOV Cliff's δ (< 0: SVO easier). * $p < .05$
Largest model per family; the same pattern holds at all sizes (560M–13B, backup).

XGLM and mGPT favour SVO, though built to broaden language coverage

BLOOM: 21 of its 28 SVO languages used in pretraining are Niger-Congo, mostly very-low-resource, which inflates the SVO group's median BPEC

SVO languages get far more training data: e.g., median SVO/SOV ratio **5×** in XGLM